SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
SPRING 2024
INTRODUCTION TO OPERATING SYSTEM (CS-222T)
Assignment # 2

Date: 24-06-2024
Due Date: 01-07-2024 Max Marks: 3

Instructions:
- Attempt all questions.

Q#1
What part of the operating system makes the decision with regards to which job will run.

Q#2
Provide three programming examples in which multithreading provides better performance than a single-threaded solution. What are two differences between user-level threads and kernel-level threads? Under what circumstances is one type better than the other?

Q#3
What resources are used when a thread is created? How do they differ from those used when a Process is created? Why should a web server not run as a single-threaded process?

Q#4
Consider a computer system that runs 5,000 jobs per month with no deadlock-prevention or Deadlock-avoidance scheme. Deadlocks occur about twice per month, and the operator must terminate and rerun about10 jobs per deadlock. Each job is worth about $2 (in CPU time), and the jobs terminated tend to be about half-done when they are aborted. A systems programmer has estimated that a deadlock-avoidance algorithm (like the banker's algorithm) could be installed in the system with an increase in the average execution time per job of about 10 percent. Since the machine currently has 30-percent idle time, all 5,000 jobs per month could still be run, although turnaround time would increase by about 20 percent on average.
a. What are your arguments for installing the deadlock-avoidance algorithm?
b. What are your arguments against installing the deadlock-avoidance algorithm?